Assignment-2

Take one Domain and draw the graph (Normal distribution) (Empirical rule).

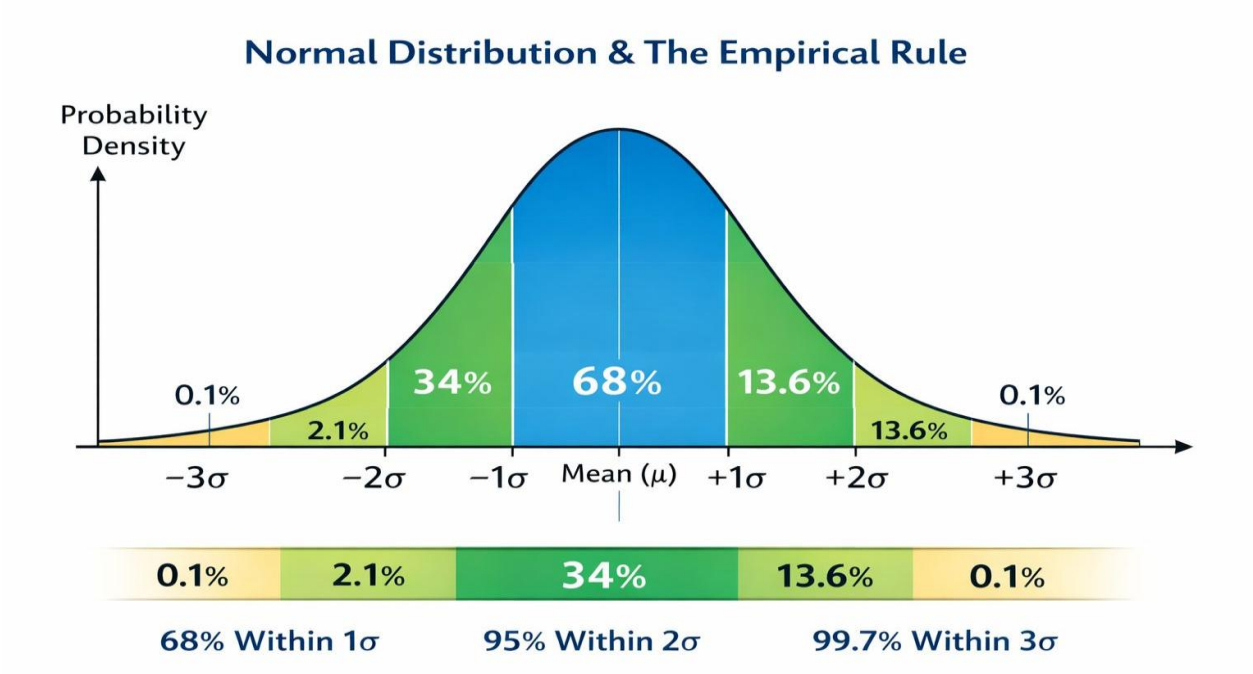
1. Introduction

The normal distribution, often called the Gaussian distribution or bell curve, represents one of the most fundamental concepts in statistics. Its ubiquity across diverse fields—from human heights and blood pressure readings to measurement errors and IQ scores—stems from a profound mathematical result known as the central limit theorem.

For this analysis, we examine the height of adult women in the United States, using parameters based on data from the Centers for Disease Control and Prevention (CDC):

· Mean (μ): 64 inches (5 feet 4 inches)

· Standard Deviation (σ): 2.5 inches



2. Mathematical Foundations

2.1 The Probability Density Function

For a normally distributed random variable X with mean μ and standard deviation σ, the probability density function is:

f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}

This equation encodes several key properties:

· The coefficient ensures the total area under the curve equals exactly 1

· The exponential term creates the characteristic bell shape

· The denominator σ controls the spread: larger σ produces wider, flatter curves

2.2 Key Properties

Symmetry and Unimodality: The distribution is perfectly symmetric about its mean, with the mean, median, and mode all coinciding at the centre. Approximately 50% of observations fall below the mean and 50% above.

Asymptotic Behaviour: The tails approach but never touch the horizontal axis, reflecting the theoretical possibility of observing values far from the mean, though such observations become exponentially less likely.

Inflection Points: The curve changes from concave to convex at exactly one standard deviation from the mean—at points $\mu \pm \sigma$.

2.3 Parameters of the Distribution

The Mean ($\mu = 64$ inches) : Determines the location of the distribution's centre. Changes to μ shift the entire curve left or right without altering its shape.

The Standard Deviation ($\sigma = 2.5$ inches) : Determines the spread or dispersion. A smaller σ would produce a taller, narrower curve with less variability; a larger σ would produce a flatter, wider curve with greater variability.

3. The Empirical Rule (68-95-99.7 Rule)

3.1 Statement of the Rule

For any normal distribution:

· Approximately 68% of observations fall within one standard deviation of the mean: $\mu \pm \sigma$

· Approximately 95% of observations fall within two standard deviations of the mean: $\mu \pm 2\sigma$

· Approximately 99.7% of observations fall within three standard deviations of the mean: $\mu \pm 3\sigma$

These percentages hold true for every normal distribution, regardless of the specific values of μ and σ.

3.2 Application to Women's Heights

One Standard Deviation Interval:

64 \pm 2.5 = [61.5, 66.5] \text{ inches}

· Lower bound: 61.5 inches (5 feet 1.5 inches)

· Upper bound: 66.5 inches (5 feet 6.5 inches)

· Contains approximately 68% of all adult women

Two Standard Deviations Interval:

64 \pm 5 = [59, 69] \text{ inches}

· Lower bound: 59 inches (4 feet 11 inches)

· Upper bound: 69 inches (5 feet 9 inches)

· Contains approximately 95% of all adult women

Three Standard Deviations Interval:

64 \pm 7.5 = [56.5, 71.5] \text{ inches}

· Lower bound: 56.5 inches (4 feet 8.5 inches)

· Upper bound: 71.5 inches (5 feet 11.5 inches)

· Contains approximately 99.7% of all adult women

3.3 Partitioning the Normal Curve

For more detailed probability calculations, the normal curve divides into eight distinct segments:

Segment Range Percentage

Left tail Below $\mu - 3\sigma$ 0.15%

Left outer $\mu - 3\sigma$ to $\mu - 2\sigma$ 2.35%

Left intermediate $\mu - 2\sigma$ to $\mu - 1\sigma$ 13.5%

Left central $\mu - 1\sigma$ to μ 34%

Right central μ to $\mu + 1\sigma$ 34%

Right intermediate $\mu + 1\sigma$ to $\mu + 2\sigma$ 13.5%

Right outer $\mu + 2\sigma$ to $\mu + 3\sigma$ 2.35%

Right tail Above $\mu + 3\sigma$ 0.15%

4. Probability Calculations

4.1 Methodology

The area under the normal curve over any interval corresponds directly to the probability of observing a value within that interval. By summing the appropriate segment percentages, we can determine probabilities for any range that aligns with standard deviation boundaries.

4.2 Worked Examples

Example 1: Percentage of women taller than 69 inches

69 inches represents $\mu + 2\sigma$ ($64 + 2\times2.5 = 69$). The area above this point consists of:

· From $\mu + 2\sigma$ to $\mu + 3\sigma$: 2.35%

· Above $\mu + 3\sigma$: 0.15%

Total: 2.35% + 0.15% = 2.5%

Only 2.5% of adult women are taller than 5 feet 9 inches.

Example 2: Percentage of women between 59 and 66.5 inches

59 inches is $\mu - 2\sigma$; 66.5 inches is $\mu + 1\sigma$. The area includes:

· From $\mu - 2\sigma$ to $\mu - 1\sigma$: 13.5%

· From $\mu - 1\sigma$ to μ: 34%

· From μ to $\mu + 1\sigma$: 34%

Total: 13.5% + 34% + 34% = 81.5%

Approximately 81.5% of women fall in this height range.

Example 3: Percentage of women shorter than 61.5 inches

61.5 inches is $\mu - 1\sigma$. The area below this point includes:

· From $\mu - 2\sigma$ to $\mu - 1\sigma$: 13.5%

· From $\mu - 3\sigma$ to $\mu - 2\sigma$: 2.35%

· Below $\mu - 3\sigma$: 0.15%

Total: 13.5% + 2.35% + 0.15% = 16%

Approximately 16% of women are shorter than 5 feet 1.5 inches.

Example 4: Percentage of women between 56.5 and 61.5 inches

56.5 inches is $\mu - 3\sigma$; 61.5 inches is $\mu - 1\sigma$. The area includes:

· From $\mu - 3\sigma$ to $\mu - 2\sigma$: 2.35%

· From $\mu - 2\sigma$ to $\mu - 1\sigma$: 13.5%

Total: 2.35% + 13.5% = 15.85%

Example 5: Percentage of women outside the range 59 to 69 inches

This range covers $\mu \pm 2\sigma$, containing 95% of the data. Therefore:

100\% - 95\% = 5\%

Due to symmetry, 2.5% are below 59 inches and 2.5% are above 69 inches.

Example 6: Percentage of women below 56.5 inches or above 71.5 inches

This asks for the combined area in both tails beyond three standard deviations:

· Below $\mu - 3\sigma$: 0.15%

· Above $\mu + 3\sigma$: 0.15%

Total: 0.15% + 0.15% = 0.3%

Only 0.3% (3 in 1000) of women fall outside the three-standard-deviation range.

5. Practical Applications

5.1 Outlier Detection

The empirical rule provides a straightforward method for identifying potential outliers. Observations falling beyond three standard deviations from the mean occur with probability less than 0.3%—fewer than three in a thousand observations.

For more extreme deviations, the probabilities become vanishingly small:

· 4σ event: Approximately 0.0063% (63 per million)

· 5σ event: Approximately 0.000057% (1 in 1.7 million)

· 6σ event: Approximately 0.0000002% (2 per billion)

5.2 Quality Control and Six Sigma

The empirical rule forms the foundation of Six Sigma methodology. The term "Six Sigma" refers to achieving process capability such that specification limits are six standard deviations from the mean, corresponding to only 3.4 defects per million opportunities.

5.3 Scientific Discovery Standards

Different fields have adopted varying standards based on the empirical rule:

· Social Sciences: Two-sigma level (95% confidence) for statistical significance

· Particle Physics: Five-sigma significance (99.99994% confidence) for discoveries like the Higgs boson

5.4 Z-Scores and Standardisation

Standardisation using z-scores allows comparison across different normal distributions:

$$z = \frac{x - \mu}{\sigma}$$

For a woman 68 inches tall (5 feet 8 inches):

$$z = \frac{68 - 64}{2.5} = 1.6$$

This indicates she is 1.6 standard deviations above the mean height. Using standard normal tables, approximately 94.5% of women are shorter.

5.5 Excel Implementation

Modern statistical analysis employs software tools like Excel:

· NORM.DIST(x, μ, σ, 1) returns cumulative probability up to value x

· 1 - NORM.DIST(x, μ, σ, 1) returns probability above value x

For women shorter than 68 inches:

$$= \text{NORM.DIST}(68, 64, 2.5, 1) \approx 0.945$$

5.6 Chebyshev's Theorem: A Non-Parametric Alternative

While the empirical rule applies specifically to normal distributions, Chebyshev's theorem provides a more general guarantee applicable to any distribution:

For any constant $k > 1$, at least $(1 - 1/k^2)$ of observations fall within k standard deviations of the mean.

For $k = 2$: at least 75% within two standard deviations (compared to 95% for normal)

For $k = 3$: at least 88.9% within three standard deviations (compared to 99.7% for normal)

6. Limitations and Considerations

6.1 The Normality Assumption

The empirical rule's accuracy depends critically on the data genuinely following a normal distribution. When data come from distributions with "fatter tails" (like financial returns), the empirical rule will overstate the concentration near the mean.

6.2 Sample Size Considerations

With small samples, the empirical rule provides only rough approximations. Small samples may deviate substantially from normality even when drawn from a perfectly normal population.

6.3 The Black Swan Problem

Nassim Nicholas Taleb criticised risk models that assume normal distributions for phenomena with extreme tail events. The 1987 stock market crash would correspond to a 36-standard-deviation event under normal assumptions—suggesting the underlying process is not well-modelled by a normal distribution.

6.4 Bimodal and Multimodal Distributions

Data sampled from populations comprising distinct subpopulations often produce bimodal distributions. For instance, heights of adult humans combined across genders produce a bimodal distribution reflecting the different means for males and females.

7. Summary and Conclusion

7.1 Key Findings

Using women's heights ($\mu = 64$ inches, $\sigma = 2.5$ inches), we have demonstrated:

The Empirical Rule:

· 68% between 61.5 and 66.5 inches

· 95% between 59 and 69 inches

· 99.7% between 56.5 and 71.5 inches

Probability Calculations:

· 2.5% of women are taller than 69 inches

· 16% are shorter than 61.5 inches

· 81.5% are between 59 and 66.5 inches

· 0.3% fall outside the 56.5 to 71.5 inch range

7.2 Implications

For Researchers: The empirical rule enables rapid assessment of whether observations are "typical" (within two standard deviations) or "unusual" (beyond two standard deviations).

For Business Professionals: Inventory management, quality control, and risk assessment benefit from understanding how variability behaves under normality.

For Healthcare Providers: Anthropometric measurements follow normal distributions, enabling identification of patients needing further evaluation.

7.3 Final Thoughts

The normal distribution's enduring importance stems from its mathematical elegance, empirical prevalence, and theoretical justification through the central limit theorem. The empirical rule—

68-95-99.7—provides an accessible entry point to understanding this fundamental concept, transforming abstract parameters into intuitive probability statements.

Whether applied to women's heights, manufacturing quality, or scientific discoveries, the normal distribution reminds us that beneath apparent randomness often lies predictable order.